

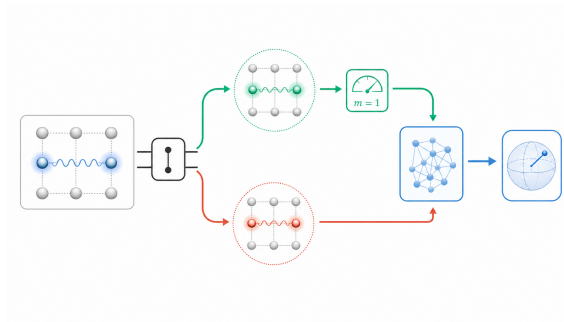
Noise-tolerant Rydberg Gates and Erasure-biased Logical Qubits

What matters beyond physical gate fidelity

Noise-Tolerant Quantum Control Optimization

July 3, 2026

One sentence



Main point

A better Rydberg gate for logical qubits is not only a higher-fidelity gate. It is a gate whose remaining errors are easier for the decoder to locate and use.

Three papers, one story

Paper	Role in the story	Takeaway for this talk
Jandura–Thompson–Pupillo, PRX Quantum 2023 [1]	Control objective for Rydberg gates	Optimize logical-qubit performance, not only Bell-state fidelity.
Ma et al., Nature 2023 [2]	Physical erasure-conversion prototype	Metastable ^{171}Yb can turn part of gate error into a mid-circuit flag.
Zhang et al., Nature Physics 2026 [3]	Logical-qubit demonstration	Erasure checks, $[[4, 2, 2]]$ decoding, and adaptive execution close the hardware-to-logical loop.

robust control → less unlocated error → more useful erasure information → better logical behavior.

Why fidelity alone misses the logical point

Same total error

$$p_{\text{tot}} = p_e + p_p.$$

$$R_e = \frac{p_e}{p_e + p_p}, \quad \eta_e = \frac{1}{1 - R_e}.$$

Different decoder input

$$\mathcal{D} : (s, m, \text{noise model}) \rightarrow R.$$

Syndrome s says which checks fired. Erasure flags m say where a physical qubit left the computational subspace.

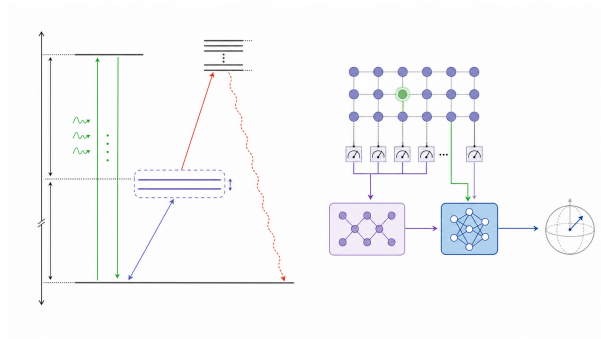
unlocated Pauli/phase
 p_p

decoder infers
location + type

located erasure
 $p_e, m_i = 1$

decoder already
knows location i

Physical story: leakage becomes a correctable erasure



Level structure.

$$|0\rangle, |1\rangle \subset {}^3P_0, \quad \mathcal{C} = \text{span}\{|0\rangle, |1\rangle\}.$$

Leakage or decay can move population to 1S_0 .

Erasure conversion. 399 nm imaging drives the cycling transition ${}^1S_0 \leftrightarrow {}^1P_1$:

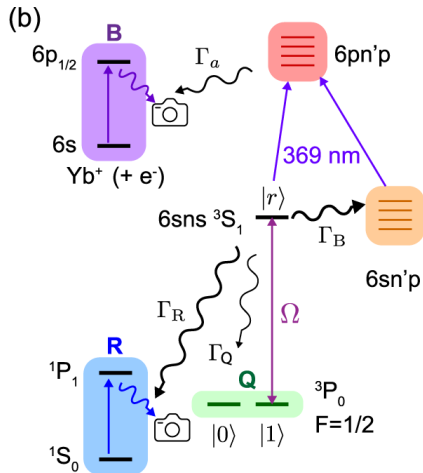
$${}^1S_0 \Rightarrow \text{bright} \Rightarrow m_i = 1, \quad {}^3P_0 \Rightarrow \text{dark}.$$

QEC use.

$$(s, m_i = 1, \text{ noise model}) \xrightarrow{\mathcal{D}} R_i.$$

The decoder knows the damaged location, so this leakage is treated as a located erasure rather than an unknown Pauli error.

Erasure conversion hardware: metastable ^{171}Yb



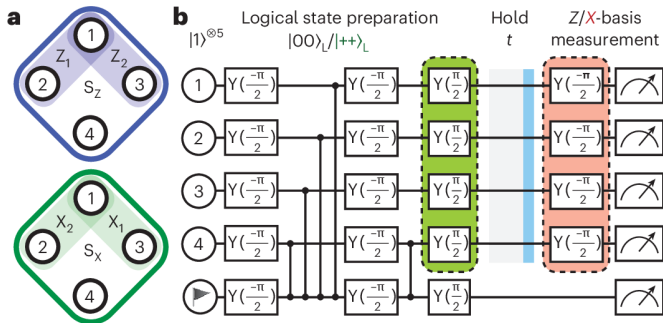
What the figure contributes

- Qubits are the two m_F states in metastable 3P_0 .
- Rydberg gates drive $|1\rangle \leftrightarrow |r\rangle$ at 302 nm .
- Decay mostly leaves $\mathcal{C}$: radiative R , blackbody B , or small return Q .
- R and B can be detected without disturbing dark 3P_0 qubits.

$$\mathcal{H} = \mathcal{C}_{3P_0} \oplus \mathcal{E}_{1S_0} \oplus \mathcal{L}_{\text{other}} \oplus \mathcal{A}_{\text{loss}}.$$

Source image: Wu et al. 2022, cropped Fig. 1b, Nature Communications [4].

QEC layer: the $[[4, 2, 2]]$ code uses erasure locations



Stabilizers:

$$S_Z = Z_1 Z_2 Z_3 Z_4, \quad S_X = X_1 X_2 X_3 X_4.$$

Distance $d = 2$ means:

$$\left\lfloor \frac{d-1}{2} \right\rfloor = 0$$

unknown Pauli errors are not generally correctable.

But one known-location erasure is usable:

$$(s, i_{\text{erased}}) \Rightarrow \text{conditional decoding.}$$

Source image: Zhang et al. 2026, cropped Fig. 3a,b [3].

Noise-robust control: where robust Rydberg pulses enter

Physical channel split

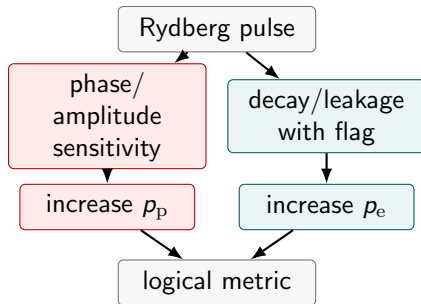
$$p_e = r p_d,$$

$$p_p \approx (1 - r)p_d + (1 - p_d)(1 - F_c).$$

p_d : Rydberg decay probability.

F_c : fidelity conditioned on returning to $\mathcal{C}$.

r : fraction of decay events converted into located erasure.



Control objective

Robust pulses can accept some extra Rydberg exposure if they suppress no-flag computational-subspace sensitivity.

How AR, DR, and ADR pulses are generated

Shared GRAPE setup. The resonant blockade CZ is written as a piecewise pulse $\Omega(t) = |\Omega_k|e^{i\phi_k}$. For each computational branch $q \in \{01, 10, 11\}$,

$$|\psi_q(\tau)\rangle = \left| \psi_q^{(0)}(\tau) \right\rangle + \sum_{\alpha} \lambda_{\alpha} \left| \psi_{q,\alpha}^{(1)}(\tau) \right\rangle + O(\lambda^2).$$

The zero-order trajectory must close and accumulate the CZ phase.

AR: amplitude robust. $\lambda_{\alpha} = \epsilon_i$ rescales the Rabi coupling. Since ϵ_i does not reverse in a second half, the pulse itself must remove first-order response:

$$J_{\text{AR}} = 1 - F + \sum_{q,i} \left\| \left| \psi_{q,\epsilon_i}^{(1)}(\tau) \right\rangle \right\|^2.$$

Cost of robustness: $\tau_R^{\text{TO}} = 2.96$, $\tau_R^{\text{AR}} = 4.74$, $\tau_R^{\text{DR}} = 5.56$, $\tau_R^{\text{ADR}} = 10.37$ in units of $1/\Omega_{\text{max}}$ [1].

DR: Doppler robust. Optimize a half pulse with half the entangling phase. Doppler shifts $\Delta_i = kv_i$ are reversed between halves by $k \rightarrow -k$ or by waiting half a trap period:

$$(I - |q\rangle\langle q|) \left| \psi_{q,\Delta_i}^{(1)}(\tau_h) \right\rangle = 0.$$

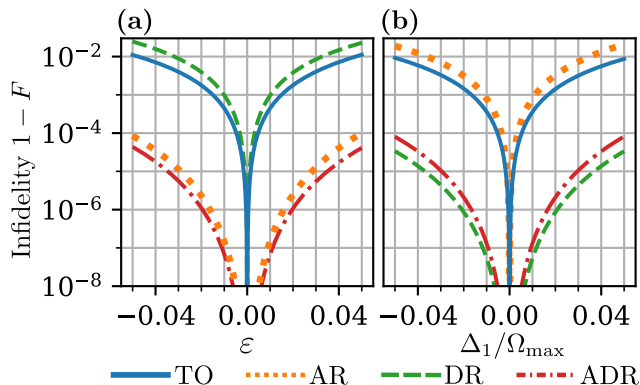
Longitudinal first-order phases are allowed because they cancel after the sign-reversed second half.

ADR: amplitude and Doppler robust. The same half-pulse variables satisfy both constraints:

$$J_{\text{ADR}} = 1 - F_h + J_{\epsilon}^{(1)} + J_{\Delta,\perp}^{(1)}.$$

It is not an AR pulse concatenated with a DR pulse; the constraints compete on one trajectory.

Robust-pulse result: flatter response to technical noise

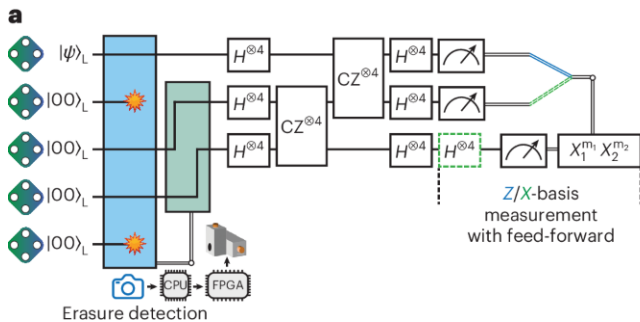


Amplitude noise. AR and ADR flatten $1 - F$ versus common Rabi-amplitude error ϵ ; TO and DR remain amplitude sensitive.

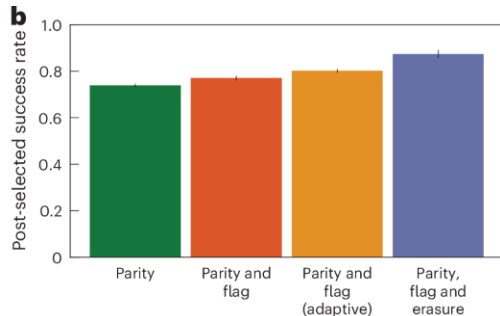
Doppler noise. DR and ADR flatten $1 - F$ versus $\Delta_1/\Omega_{\max}$; ADR is the joint robust choice when both noise sources matter.

Source image: Jandura et al. 2023, cropped Fig. 2a,b, PRX Quantum [1].

Overall result: erasure flags become useful logical information



Protocol ladder. The experiment compares parity-only operation, adding flag information, adaptive feed-forward, and finally adaptive decoding with erasure records.



Main effect. The post-selected success rate increases across this ladder, showing that erasure conversion is not only a diagnostic: it supplies location information that can be used during logical execution.

Source image: Zhang et al. 2026, Fig. 4, Nature Physics [3].

The deck's compressed takeaway

- ① The logical layer cares about the pair (p_e, p_p) , not only $p_e + p_p$.
- ② Metastable ^{171}Yb gives a useful physical interface: qubits are dark, some leakage/decay is bright.
- ③ $[[4, 2, 2]]$ is intentionally small: it makes the value of one known erased location visible.
- ④ Adaptive logical circuits are the real milestone: erasure flags are used during execution, not only after the experiment.
- ⑤ For this repository, the natural next simulation target is an erasure-aware physical-to-logical score for shaped ^{171}Yb CZ pulses.

References I

- [1] Sven Jandura, Jeff D. Thompson, and Guido Pupillo. “Optimizing Rydberg Gates for Logical-Qubit Performance”. In: *PRX Quantum* 4.2 (June 5, 2023), p. 020336. ISSN: 2691-3399. DOI: 10.1103/PRXQuantum.4.020336. URL: <https://link.aps.org/doi/10.1103/PRXQuantum.4.020336> (visited on 06/22/2026).
- [2] Shuo Ma et al. “High-Fidelity Gates with Mid-Circuit Erasure Conversion in a Metastable Neutral Atom Qubit”. In: *Nature* 622.7982 (Oct. 12, 2023), pp. 279–284. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-023-06438-1. arXiv: 2305.05493 [quant-ph]. URL: <http://arxiv.org/abs/2305.05493> (visited on 05/06/2026).
- [3] Bichen Zhang et al. “Logical Qubits with Erasure Conversion Using Metastable Neutral Atoms”. In: *Nature Physics* 22.6 (June 2026), pp. 910–916. ISSN: 1745-2481. DOI: 10.1038/s41567-026-03309-0. URL: <https://www.nature.com/articles/s41567-026-03309-0> (visited on 06/22/2026).

- [4] Yue Wu et al. “Erasure Conversion for Fault-Tolerant Quantum Computing in Alkaline Earth Rydberg Atom Arrays”. In: *Nature Communications* 13.1 (Aug. 9, 2022), p. 4657. ISSN: 2041-1723. DOI: 10.1038/s41467-022-32094-6. URL: <https://www.nature.com/articles/s41467-022-32094-6> (visited on 04/16/2026).